

GenomeLens Deterministic Benchmark v1

Frozen public-data technical proof of concept for genomic health-economic decision analysis

GenomeLens is a research platform for translating genomic evidence into decision-relevant health economics: what decision a finding could change, what health and cost consequences could follow, how uncertainty should be represented, and when the evidence is insufficient to assign a defensible economic value.

Deterministic Benchmark v1 evaluates that decision layer across **36 public benchmark genomes**: 30 primary public genomes plus 6 reference/parent genomes. The benchmark is designed to test reproducibility, economic routing, arithmetic integrity, report parity, explicit zero/missingness semantics, and public-report integrity. It is **not** a clinical validation, diagnostic-performance study, population-effectiveness estimate, or regulatory claim.

Frozen v1 results

Validation layer	Frozen v1 result
Cohort completion	36 / 36
Engine	4eaf729
Tooling	a9b3d3a
Full clean-tree suite	7,723 passed, 26 skipped, 0 failed
Release gates	9 / 9
Parity failures	0
Zero audit	0 defects across 1,399 displayed zeros
Arithmetic audit	0 discrepancies
Lost-content audit	0 downgraded rows
Real-row assertions	301 evaluated, 0 failures
PDF semantic checks	36 / 36
Forbidden-string scan	0 hits across 144 rendered PDFs
Analyst challenge review	278 / 278 answered; no P0/P1 finding

Visual-gate disposition

The visual gate flagged **49 of 9,482 rendered pages**, all in the full-genome **report.pdf**. The economics PDFs were clean. The flags consisted of a bounding-box false positive plus minor ancestry-label overlap/clipping and were accepted as a **non-material P2 presentation exception**. This is not represented as an unconditional visual pass.

What GenomeLens demonstrates

GenomeLens separates the genomic evidence state from the economic disposition of a finding. A biologically supported result does not automatically receive a dollar value. Depending on the evidence and decision context, the economic result can be modeled, conditional, unresolved, explicitly zero, unquantified, or refused.

The public benchmark demonstrates two distinct analysis modes:

Individual Genomic Economics

one genome -> evidence-qualified finding -> decision/comparator
-> health consequence -> cost/QALY effect -> ICER/NMB/uncertainty

Reproductive & Carrier Economics

Parent A + Parent B -> inheritance compatibility -> offspring
probability -> reproductive decision -> expected health/economic
consequence

Pair-level reproductive value is treated as a distinct estimand and is not silently added to either parent's personal-health value.

Economic outputs

Where the evidence chain supports quantification, GenomeLens can represent:

- incremental healthcare cost;
- QALY impact;
- ICER;
- net monetary benefit (NMB);
- context-specific or if/when-decision value;
- sensitivity and scenario analysis;
- value-of-information concepts;
- reproductive / family-planning economic consequences;
- explicit true-zero, unresolved, unquantified, and refusal states.

A central benchmark rule is **missing is not zero**: a pathway that lacks the technical resolution, donor context, intervention evidence, or separable economic terms required for a defensible estimate is not silently rendered as \$0.

What the benchmark establishes

- reproducible execution across the intended public-genome cohort;
- consistent routing between genomic evidence states and economic dispositions;
- arithmetic reconciliation for recomputable economic outputs;
- explicit handling of true zero, missing context, conditionality, uncertainty, and refusal;
- report-to-structured-result consistency across public economics PDFs;

- pair-level reproductive decision analysis as a separate unit of analysis;
- resistance to unsupported external analyst suggestions entering the deterministic economics layer.

What it does not establish

- diagnostic sensitivity or specificity of whole-genome sequencing;
- population prevalence or representativeness;
- clinical efficacy for an individual patient;
- a single cash value of a genome;
- personalized medical or reproductive advice;
- regulatory clearance or medical-device validation.

Disclosed release limitations

Two validation activities are explicitly recorded as future work rather than represented as completed:

1. Mutation harnesses were not rerun because their source anchors were stale.
2. No separate full manual adversarial-review pass was completed across every final artifact; one Lynch report was read end-to-end.

In addition, 20 economic rows were kept explicitly classified as not independently recomputable from separable delta-QALY / delta-cost terms rather than being back-filled with fabricated components.

Reproducibility record

- **Benchmark:** GenomeLens Deterministic Benchmark v1
- **Engine commit:** 4eaf729
- **Tooling commit:** a9b3d3a
- **Cohort:** 30 primary public genomes + 6 reference/parent genomes
- Frozen benchmark artifacts were archived and integrity-checked locally.
Internal benchmark packages and raw run outputs are not publicly distributed.
- **Personal/private inputs:** excluded from the public benchmark artifact

Public communication boundary

The public benchmark is intended to demonstrate the research framework, economic outputs, validation evidence, and limitations without disclosing implementation details sufficient to reconstruct the proprietary engine. Public materials should therefore avoid publishing proprietary route tables, internal parameter registries, private model logic, raw private genomes, personal bloodwork, credentials, or other protected implementation material.

Status: DETERMINISTIC BENCHMARK v1 - FROZEN

Public-data technical proof of concept. Not clinical validation.